

Joana Louro

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Profile

Artificial Intelligence Master's student at the FEUP, with hands-on experience in algorithms, autonomous systems and optimization. Led a team to the finals of the 2026 Bosch Future Mobility Challenge, achieving 6th place worldwide. I am a competitive volleyball athlete, driven by the same pursuit of excellence in both sport and engineering.

Experience

Teaching Assistant – Algorithm Design Course March 2026 – July 2026
Faculty of Engineering University of Porto (FEUP) *Porto, Portugal*
- Reviewed and validated programming-based group projects for the Algorithm Design course.
- Developed model solutions and evaluation datasets for project assessment.
- Implemented automated correction tools to support project evaluation.

FSFEUP – Formula Student FEUP Porto, Portugal
Data Infrastructure Subsystem Member – Autonomous Systems Department *March 2025 – August 2025*
- Developed a ROS node to extract rosbag data and populate a PostgreSQL database for vehicle performance analysis.
- Integrated Apache Superset dashboards to visualize vehicle metrics and performance indicators.
- Automated rosbag deployment and data ingestion through server-side scripts.

Head of Path Planning Subsystem – Autonomous Systems Department *September 2025 – September 2026*
- Refactored the path-planning codebase and optimized the mean-path generation.
- Used optimization-based planning to generate high-performance racing trajectories while enforcing safety constraints.
- Implemented velocity profiling to determine feasible speeds based on vehicle dynamics and physical constraints.
- Developed adaptive velocity planning that segments straights and curves and adjusts speeds based on tracking deviations.

Awards and Competitions

Bosch Future Mobility Challenge 2026 | *Team Leader, 6th place worldwide*
- Led a team to the finals of the 2026 Bosch Future Mobility Challenge, achieving **6th place worldwide** among 141 applicant teams and ranking as the competition's **best new team**.
- Developed an Extended Kalman Filter (EKF) for vehicle state estimation through asynchronous fusion of GPS, IMU, and wheel encoder data.
- Designed and implemented real-time autonomous vehicle control systems, integrating behavioral decision logic, Pure Pursuit steering, and reliable path tracking in looped-track environments.

Europe Women's Programming Contest (EWPC) 2026 | *Finalist*
- Finalist in the 2026 EWPC, an ICPC competition. Only the **top 40** female competitors in **Europe** were selected.

Formula Student Competitions | *FS Portugal and FS Switzerland*
- Achieved 1st place in the Junior Team category at Formula Student Portugal and Formula Student Switzerland.
- Presented and technically defended FSFEUP's autonomous system before the competition judges.

Volleyball 2025 | *Federated and University Athlete*
- 2nd place in the National University Championship and 4th place in the sub21 First National Division.

Education

Faculty of Engineering University of Porto (FEUP) Porto, Portugal
Master in Artificial Intelligence *September 2026 – Current*
Faculty of Engineering University of Porto (FEUP) Porto, Portugal
Bachelor's Degree in Informatics and Computing Engineering | Final grade: 17/20 *September 2023 – July 2026*
Cambridge English
Advanced (CAE) - Certificate of Advanced English *July 2023*

Technical Skills

Languages: C/C++, Python, Java, SQL, Rust
Technologies/Tools: Git, Docker, PostgreSQL, TimescaleDB, Apache Superse, Foxglove, ROS 2